



Building Resilience and Enabling Adaptation to Climate Change for Host and Refugee Communities: An Assessment of SHARPE

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Contents

Executive summary	4
Introduction	5
Methodology.	5
The Ethiopian Context	7
Climate Change Trends and Projections	8
Climate Change Projections for Ethiopia	9
Economic impacts	9
Regional Variation	10
Climate Risk and Vulnerability in Somali and Gambella Regions	11
Vulnerability of Refugees	14
Strengthening Host and Refugee Populations in Ethiopia – SHARPE	16
Climate Risk Assessment	16
Climate Risk Assessment of SHARPE Interventions	18
Climate Risk Summary for SHARPE	25
UK International Climate Finance (ICF): Contribution to adaptation and mitigation	27
Putting Climate Risk and Adaptation at the Centre of Programmes.	30
Conclusions.	30
Annex 1: Glossary	31

Executive summary

This report presents the findings of an assessment of the **Strengthening Host and Refugees Populations in Ethiopia (SHARPE)** programme on the contribution of its interventions in strengthening the climate resilience of refugee and host communities. Ethiopia is one of most vulnerable countries in the world to climate change. Within Ethiopia, refugee settlements are in the regions experiencing the highest climate risks from rising temperature, drought and flooding. SHARPE works in refugee communities in two high-risk regions, Gambella and Somali. SHARPE aims to catalyse transformational system change in how donors, governments, and the private sector respond to the protracted refugee contexts in Ethiopia by using a market systems development approach to strengthen the economies of the target communities in Dollo Ado and Jijiga (Somali region) and Gambella. The programme works to increase refugee self-reliance and generate economic opportunities for host and refugee communities through the piloting and scaling of interventions. The programme also aims to serve as a learning platform for the wider international and Ethiopian development community to share lessons in the application of market-oriented systemic approaches in humanitarian settings.

With the impact of climate change being increasingly experienced in vulnerable countries such as Ethiopia, there is mounting concern that climate change will undermine development gains. In response to these concerns, there is a call for climate risk and climate action to be integrated into all relevant sectors and programmes. The purpose of this assessment was to apply a climate lens to review SHARPE's interventions and activities and consider the climate risk of each intervention (e.g. the interventions may currently be economically sustainable, but would they be sustainable in a hotter, drier, and flooding environment), what could be done to strengthen adaptive capacity, and how these adaptation measures could be integrated into its interventions. Although SHARPE does not receive UK ICF funding, the ICF guidelines (KPI.1 and KPI.2.1) were used to determine SHARPE's contribution to climate adaptation and mitigation.

The assessment found that climate risk and climate resilience were not explicitly highlighted or integrated into SHARPE's programme design. This lack of attention to and integration of climate risk is common in refugee-oriented and economic development programmes; while climate change may be mentioned in the background context, it is not directly addressed or highlighted in objectives, theory of change, or targets. However, even if not explicitly linked to climate risk and resilience, programme interventions and activities can make a significant contribution to strengthening climate resilience.

This is found in SHARPE's programme and interventions. While not explicitly citing climate adaptation in its programme design, SHARPE's MSD interventions are increasing resilience by strengthening the ability of communities to adapt to climate stresses and variability. Measures to strengthen the climate resilience of the most at risk interventions (e.g. poultry hub, goat rearing) are both pragmatic and cost-effective. Energy interventions are facing minimal climate risks and are providing clean energy that was not previously available. Refugee and host communities are benefiting not only from increased income, but also from a diversification of livelihood opportunities which is a key factor in adaptation to climate (and economic) shocks. Using the ICF criteria it was found SHARPE had supported (to date) **181,778** people to better adapt to the effects of climate change and **14,266** people to have improved access to clean energy.

Looking to the future, the assessment findings highlight the importance of including climate risk from the very beginning in the planning and design of programmes for protracted refugee communities to better enable the long-term sustainability of initiatives in hotter, drier yet flood prone regions. In its role as a learning platform, SHARPE can share these findings and the steps it has undertaken to integrate climate risk into its own interventions. It can also share that a market-oriented systemic approach that aims to increase refugee self-reliance can make meaningful contributions to climate adaptation and mitigation and can play a pivotal role in strengthening resilience to climate shocks now and in the future.

Introduction

This report presents the findings of an assessment of the **FCDO funded Strengthening Host and Refugees Populations in Ethiopia (SHARPE)** programme on the contribution of its interventions in strengthening the climate resilience of refugee and host communities. The purpose of this assessment was to consider within the context of Ethiopia's climate change vulnerability, particularly in the Somali and Gambella regions in which there are protracted refugee settlements, the climate risks to SHARPE's interventions, what could be done to strengthen adaptive capacity, and how these adaptation measures could be integrated into its interventions. The assessment also considered SHARPE's contribution to the climate adaptation and mitigation of the refugee and host communities in which it works. Looking to the future, this report suggests actions in current and future programmes to strengthen the climate resilience of refugee and other vulnerable communities.

Methodology

The primary purpose of the assessment was to consider the current and future climate risks to the livelihoods of host and refugee communities in Ethiopia and review the extent to which SHARPE supported business innovations had strengthened climate resilience. It assessed the risks posed by climate change to a range of business innovations introduced by SHARPE, with a specific focus on Dollo Ado in the south of the Somali region, and identified further adaptations that could be made to these business models to lower climate risks and increase adaptive capacity of both host and refugee communities.

The assessment considered:

- Future climate change scenarios for the refugee hosting regions and potential risks to livelihood strategies that are common to the regions.
- Whether interventions which are currently perceived as economically sustainable (e.g. continuing after SHARPE ended) will encounter future climate risk. Interventions may be sustainable in the current context but would they be sustainable in a hotter, drier and wetter environment?
- What would be needed to strengthen the adaptive capacity of the intervention to drier conditions/drought, rising temperatures, heavy precipitation events (HPE) and flooding?
- What pragmatic, on-the-ground climate-proofing measures could be used that build on SHARPE's successes and lessons learned?
- How could these measure be integrated into current interventions and highlights for future programmes and activities? What additional resources would be needed? How could these resources be accessed?
- How is SHARPE contributing to climate adaptation and mitigation? How can interventions which can withstand climate risk be replicated or scaled?
- What recommendations can be made for building resilience and enabling adaptation for refugees and host communities?
- What recommendations can be made to better design future interventions and enable them to withstand climate risk?

This assessment was conducted in May-June 2024 and included a visit to Dollo Ado in the Somali region where approximately 210,000 refugees have lived in five large camps since 2010/11. Prior to the field visit, there was extensive research on the current and future climate risks of Ethiopia. A Climate Risk Assessment for Ethiopia was prepared focusing on the climate risks to livelihoods in the Somali and Gambella regions which is where the SHARPE programme is focused and which together host more than 50% of the total refugee population in Ethiopia. The Climate Risk Assessment was shared with the SHARPE staff and provided the context for the discussion of the vulnerability of interventions to current and projected climate risks and the modifications that could be made to increase resilience to higher temperatures, drought and floods.

Interviews were conducted with the staff of SHARPE, UNHCR (Addis and in Dollo Ado), FCDO, and with programme partners from the private sector. There was a research visit to Dollo Ado in the south of the Somali region that provided an opportunity to meet with UNHCR staff, conduct interviews with programme participants in different refugee settlements and in Dollo Ado town, observe the refugee settlements and the condition of the roads and infrastructure.¹

The interviews with programme participants were informal and free flowing conducted at their place of business or home (e.g. poultry and goat sheds within the household compound), with questions focusing on:

- Background and length of time in the area;
- Impact of climate change (e.g. temperature, drought and recent flooding);
- Perceptions of future climate risk with particular attention to the specific intervention (e.g. impact on Poultry production with the poultry hub leader, etc.);
- Benefits from the SHARPE intervention and difficulties or set-backs experienced;
- Livelihood strategy (e.g. sources of income and social capital/network) and ambitions.

The findings from the interviews are included in the assessment (see below). Among the climate related takeaways from the interviews was the awareness by the programme participants on the ‘knock on’ effect of climate events such as the recent flooding, which washed away bridges and roads, hindering travel and the flow of goods, which in turn resulted in higher prices. From the interviews the importance of social capital, especially for refugees, frequently emerged. Unlike in Gambella, where the host and refugee communities differ in ethnicity and livelihoods and have a history of conflict and tension, refugee-host relations in the Dollo Ado are positive with a shared ‘Somali’ identity and culture, language and religion. This offers an opportunity for socio-economic collaboration and integration, providing refugees with networks and opportunities that span not only the local areas but across the transnational border.

It is important to note a limitation to the assessment in that field work was only conducted in one of the three refugee hosting regions where SHARPE works, in the southern region of Dollo Ado. As such, the interventions that were assessed against future climate change scenarios were those that have been implemented in Dollo Ado. While these interventions (poultry farming; goat trading; solar energy; financial inclusion) are also being implemented in Jijiga and Gambella, the climate change risks and future scenarios are different for those two regions. There are also interventions in climate smart agriculture (improved vegetable seeds; hermetically sealed post-harvest storage bags) being implemented by SHARPE in Gambella that were not included in this assessment.



¹ Due to security concerns field visits were not undertaken in Gambella. The examples presented in this report are drawn from the field visit to Dollo Ado, interviews, and the reports of UNHCR and SHARPE.

The Ethiopian Context

Ethiopia is a landlocked country in the Horn of Africa. More than half of the country is more 1,500 meters above sea level, but there are also low-lying areas of tropical savannah and desert. Ethiopia has been referred to as the 'water tower of Africa' – it has 12 water basins, 22 major lakes, and an estimated groundwater potential of about 2.6 billion m³.² With a population of almost 130 million (March 2024), it has the second largest population in Africa after Nigeria.³ It is also one of the fastest growing with its population projected to reach over 213 million in 2050,⁴ reflecting its high female fertility rate of 4.2 children.⁵ Ethiopia ranks 75 out of 148 countries in Gender Gap Index (2023),⁶ comparable to Kenya (77) and Uganda (78).⁷

Ranked 176 out of 193 countries on the Human Development Index (2022), Ethiopia is one of the poorest countries in world.⁸ Classified as a low-income and least developed country, Ethiopia aspires to achieve lower-middle-income status by 2025. Ethiopia's economy is the sixth largest in Africa and compared with other countries in the region its economy has had a strong growth rate. Ethiopia's GDP annual growth has averaged 5.97 percent from 1981 until 2023, and the economy is expected to grow by 7.9% in 2023/24. There is concern that regional political instability, internal conflicts, and slow progress on economic structural reforms risk undermining the economic and social development progress that has been achieved.

While development patterns are shifting, agriculture remains the largest sector, accounting for more than 80 percent of the exports, employing about 64 percent of the population – most of whom are smallholders – and making up 38 percent of GDP. Agricultural production is primarily rainfed with only about 5% of the national crop land irrigated. Ethiopia also has the largest livestock population in Africa. In the drier lowlands, livelihoods are generally based on raising livestock instead of farming.⁹ Ethiopia has historically been vulnerable to food insecurity and has often relied on food aid. Production of key crops is not increasing fast enough to keep up with the country's high population growth.

Ethiopia has an open door policy for refugees and allows humanitarian access and protection to those seeking asylum. Changes in Ethiopia's refugee commitments and policies are in the process of being implemented which intend to grant a wide range of rights to refugees including education, mobility (out of refugee camp travel), documentation, right to work and access to basic social services.

Currently, Ethiopia is home to the second largest refugee population in Africa with over million refugees (1,051,000, May 2024)¹⁰ primarily from Somalia, South Sudan and Eritrea, many of whom have been in Ethiopia for more than 10 years and show no sign of leaving.^{11,12} While Ethiopia is now moving to a settlement and integration approach, historically the approach has been to house refugees in camps, most of which are located in border regions such as the Gambella and Somali regions, which are also some of the poorest and most underdeveloped regions of Ethiopia. These regions are highly vulnerable to climate change with limited income generating opportunities and frequent conflict caused by competition over land and resources.

2 Degefu, Dagmawi & He, Weijun & Zhao, Jian. (2015). Hydropower for sustainable water and energy development in Ethiopia. Sustainable Water Resources Management. 1. 10.1007/s40899-015-0029-0. https://www.researchgate.net/publication/282352906_Hydropower_for_sustainable_water_and_energy_development_in_Ethiopia

3 World Population Review – Ethiopia. <http://worldpopulationreview.com/countries/ethiopia-population/>

4 https://www.ined.fr/en/everything_about_population/data/world-projections/projections-by-countries/

5 <https://data.worldbank.org/indicator/SP.DYN.TFRT.IN?locations=ET>

6 Global Gender Gap Report (2023) https://www3.weforum.org/docs/WEF_GGGR_2023.pdf. The GGGI annually benchmarks gender parity across four key dimensions (Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment).

7 Due to lack of data, the GGGI did not include Eritrea, Somalia, Sudan, or South Sudan.

8 UNDP (2022) Human Development Reports (Ethiopia) <http://hdr.undp.org/data-center/specific-country-data#/countries/ETH>

9 BMZ, PIK, GIZ and KfW (German Federal Ministry for Economic Cooperation and Development, Potsdam Institute for Climate Impact Research, Deutsche Gesellschaft für Internationale Zusammenarbeit, and Kreditanstalt für Wiederaufbau). n.d. Climate Risk Profile: Ethiopia. https://www.pik-potsdam.de/en/institute/departments/climate-resilience/projects/project-pages/agrica/climate-risk-profile_ethiopia_en

10 <https://data.unhcr.org/en/country/eth>

11 <https://data.unhcr.org/en/country/eth>

12 'In addition to a large refugee population, Ethiopia has more than 3,779,000 internally displaced people (IDPs). Rapid urban expansion, ongoing conflicts within Ethiopia and high levels of vulnerability to ongoing drought and seasonal floods continue to generate new displacements yearly.' <https://www.unrefugees.org/news/ethiopia-refugee-crisis-explained/>

Climate Change

Climate change poses a huge challenge to Ethiopia and its people. Almost 90% of its surface is vulnerable to severe or extreme climate stresses. One of the world's most drought prone countries, drought, often linked closely with El Niño events, has been and continues to be the most damaging climate stressor in the country. Ethiopia is experiencing increasingly unpredictable rains, and in some years the complete failure of seasonal rains, occurrences that are linked to climate change. In Ethiopia drought and high temperature extremes are the main drivers hindering crop and livestock productivity, increasing household level food and nutrition insecurity, poverty and inequality and decreasing agricultural employment, which in turn are correlated with a higher likelihood and intensity of conflict. By threatening food security, climate change will also likely worsen conflict not only within Ethiopia but within and across the Horn of Africa countries. In the region climate change accompanied by environmental pressures have severe cascading effects on peace and human security, ranging from climate migration and displacement to food insecurity and conflict over diminishing natural resources.¹³

According to the ND-GAIN index¹⁴ (2023) for climate vulnerability, Ethiopia ranks 163 out of 185 countries, ranking the 24th most vulnerable country and the 156th most ready country meaning that it is vulnerable to, yet largely unready to combat climate change effects.¹⁵ Although globally one of the most vulnerable countries, Ethiopia is a low GHG emitter, contributing only 0.36% of global GHG emissions (2022).¹⁶

Climate Change Trends and Projections

In Ethiopia climate is largely influenced by elevation: the highland areas exhibit mean annual temperatures of 11.9°C (Amhara), while the lowlands exhibit mean annual temperatures of up to 30.8°C. Annual precipitation ranges from 100mm in the lowland steppe and desert to over 1900mm in the highlands. There is a single rainy season in the Ethiopian highland centre, north and west, and two rainy seasons in the lowland south and east regions.

Trends

Since 1960 Ethiopia's climate trends include:^{17,18}

- Annual average **increase in temperature** by 1.7°C (1960–2020), with most noticeable increases in already dry and hot areas of the country and in the July–September season.
- An overall **decline in precipitation** levels in the last three to four decades, with regional variability and significant year-to-year volatility. Southern and western Ethiopia experienced the most significant precipitation decline. The Gambella Region saw an average annual precipitation decrease of over 130mm/decade (approx. 10 percent) from 1971–2020. The Belg (Feb–May) rains are increasingly unpredictable.
- Changes in temperature and rainfall have **increased the frequency and severity of extreme events**.
- **Warming has exacerbated droughts**, and desertification in the lowlands is expanding.

13 L. Destrijcker et al. (2023) Weathering Risk: Climate, Peace and Security Study: Somali Region, Ethiopia. Adelphi Research. https://adelphi.de/system/files/document/climate_peace_security_study_somali_region_ethiopia.pdf

14 The widely used ND Gain Index summarizes a country's vulnerability to climate change and other global challenges in combination with readiness to improve resilience. <https://gain.nd.edu/our-work/country-index/rankings/>.

15 Vulnerability measures the exposure, sensitivity, and ability to cope with climate related hazards by accounting for the overall status of food, water, environment, health, and infrastructure within a country. Readiness targets those portions of the economy, governance and society that affect the speed and efficiency of adaptation.

16 Edgar (Emissions Database for Global Atmospheric Research] (2023) https://edgar.jrc.ec.europa.eu/report_2023#:~:text=China%2C%20the%20United%20States%2C%20India,61.6%25%20of%20global%20GHG%20emissions.

17 USAID (2024) Climate Change Risk Profile Ethiopia. <https://www.climatelinks.org/resources/climate-risk-profile-ethiopia>

18 World Bank Group (2020) Climate Risk Country Profile: Ethiopia. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15463A-WB_Ethiopia%20Country%20Profile-WEB.pdf

Climate Change Projections for Ethiopia

Climate projections are for the current trends to continue with rising temperatures, uncertain rainfall, more hot days and fewer cold days and nights and an increase in extreme events, especially droughts.

Temperature

- Across all emission scenarios,¹⁹ temperatures will continue to increase for Ethiopia throughout the century.
- Under the medium/high emissions scenarios, projected mean temperature changes will rise by 1.5°C by 2030, 1.8°C by the 2050s, and up to 3.7°C by the 2080s, with higher temperature increases concentrated in eastern Ethiopia.
- Temperature increase is also expected to result in more intense heat waves and higher rates of evapotranspiration.²⁰

Rainfall

- Rainfall projections vary significantly with climate models having high levels of uncertainty and vary spatially. What is agreed is that future dry and wet periods are likely to become **more extreme**, with severe droughts in one year and heavy flooding with erosion and landslides the next. Projections show heavy rainfall events are likely to increase by 20% by the end of the century and may account for a greater proportion of total precipitation.

Figure 1: Climate projections



Projected increase in temperature of 1.8°C by 2025



Erratic rainfall, increased unpredictability of seasonal rains, more frequent heavy precipitation events



More frequent and severe dry spells and drought events

Economic impacts

Climate change is expected to have significant economic effects on the Ethiopian economy, and on the agriculture sector in particular, with the economic impacts depending largely on the extent of annual weather variability and extremes such as drought. On national average, climate explains 55–89% of the year-to-year yield variability.²¹ It is estimated that major droughts have reduced the country's GDP by 1% to 4%, and rain-induced soil erosion has been estimated to reduce GDP by approximately an additional 1%.

Estimates suggest climate change may reduce Ethiopia's GDP by as much as 10% by 2045, largely through drought-induced impacts on agricultural productivity.²² However, it has been suggested that the projected impacts of climate change on growth will be lower if macroeconomic stabilisation and structural reforms are undertaken. For agriculture this would include expanding access to finance and insurance options.²³

19 The climate models that are commonly used are the Representative Concentration Pathways (RCPs) which represent future global warming scenarios which reflect different future emissions. The RCP 2.6 assumes very low future emissions, RCP 4.5 assumes an intermediate scenario, while RCP 8.5 assumes very high future emissions (often referred to as the business as usual' scenario in which fossil fuel use will further increase, and emissions continue to rise throughout the century, etc.) Currently the high emission scenario is being recognised as the more likely scenario due to rising emissions and the lack of adequate action in climate mitigation.

20 USAID (2024) Climate Change Risk Profile Ethiopia. <https://www.climatelinks.org/resources/climate-risk-profile-ethiopia>

21 BMZ, PIK, GIZ and KfW (German Federal Ministry for Economic Cooperation and Development, Potsdam Institute for Climate Impact Research, Deutsche Gesellschaft für Internationale Zusammenarbeit, and Kreditanstalt für Wiederaufbau). n.d. Climate Risk Profile: Ethiopia. https://www.pik-potsdam.de/en/institute/departments/climate-resilience/projects/project-pages/agrica/climate-risk-profile_ethiopia_en

22 World Bank Group (2020) Climate Risk Country Profile: Ethiopia. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-05/15463A-WB_Ethiopia%20Country%20Profile-WEB.pdf

23 World Bank Group (2023) Country Climate and Development Report (CCDR). <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099022224202134449/p17952512670c208619205179e2783cefc5>

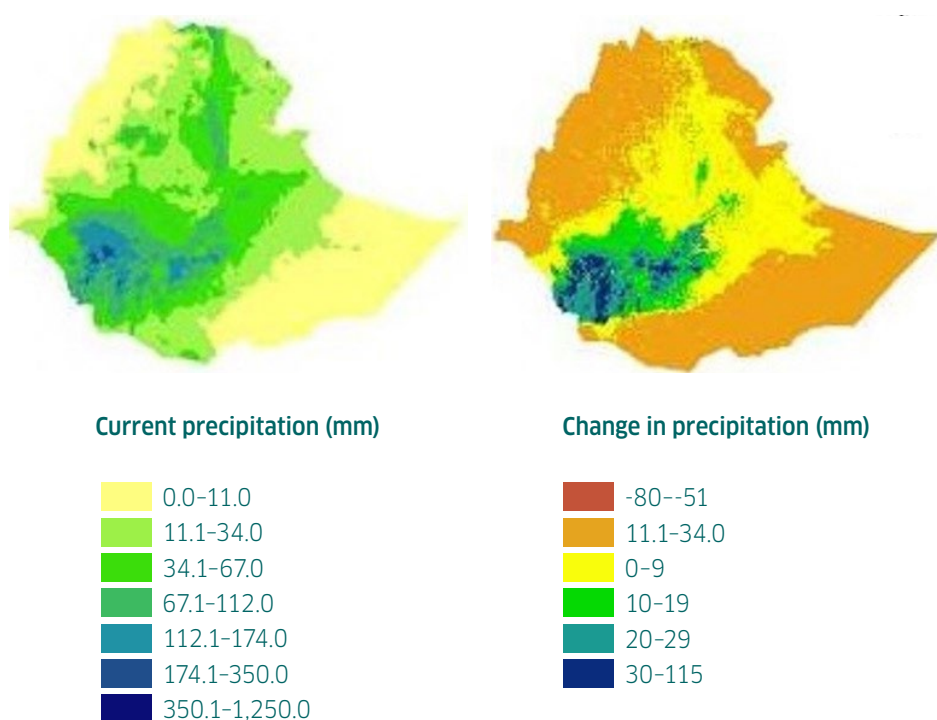
For households, through its adverse effect on food security and economic development climate change has strong links with poverty and hunger. Poverty and hunger also decrease people's resilience and adaptive capacity to climate change effects.

Regional Variation

Reflecting topographical differences across the country, there are large regional differences to climate risk vulnerability with precipitation and water availability strongly dependent on elevation. While more attention has been given by the government to the highland regions, the lowland regions are projected to be at greater climate risk with prolonged droughts and increasing temperatures affecting the agropastoral production system in the lowland regions. The highlands are projected to experience extreme events such as dry spells coupled with irregular and intense rainfall.

The largest precipitation decrease during the driest quarter of the year will be in regions that are already very dry (see Figure 2). Water supply will be affected throughout the country, but most severely in areas that are already water-scarce, such as densely populated parts of the Rift Valley and eastern pastoralist zones. Under high emissions scenarios, per capita water availability will decline by 65% by 2080 (mostly due to population growth).²⁴

Figure 2: Current precipitation during the driest season (left) and projected change (right)



Source: Hopping, K.; Wann, G. (2009): Human Population Response to Climate Change in Ethiopia. Colorado State University. http://gisedu.colostate.edu/webcontent/nr505/ethiopia2009/nr505_fall09_final_Hopping_Wann/Results.html#PopulationDensity

There is concern that the regional differences in the impacts on agriculture and livestock production will further increase regional inequality and exacerbate tensions by increasing migration and inter-group conflict over resources like water and pastureland, especially in the highly vulnerable lowland areas. These climate driven pressures will add to the strains experienced by the large number of people forcibly displaced by years of political conflict and drought.²⁵

- 24 BMZ, PIK, GIZ and KfW (German Federal Ministry for Economic Cooperation and Development, Potsdam Institute for Climate Impact Research, Deutsche Gesellschaft für Internationale Zusammenarbeit, and Kreditanstalt für Wiederaufbau). n.d. Climate Risk Profile: Ethiopia. https://www.pik-potsdam.de/en/institute/departments/climate-resilience/projects/project-pages/agrica/climate-risk-profile_ethiopia_en
- 25 World Bank Group (2023) Country Climate and Development Report (CCDR). <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099022224202134449/p17952512670c208619205179e2783cefc5>

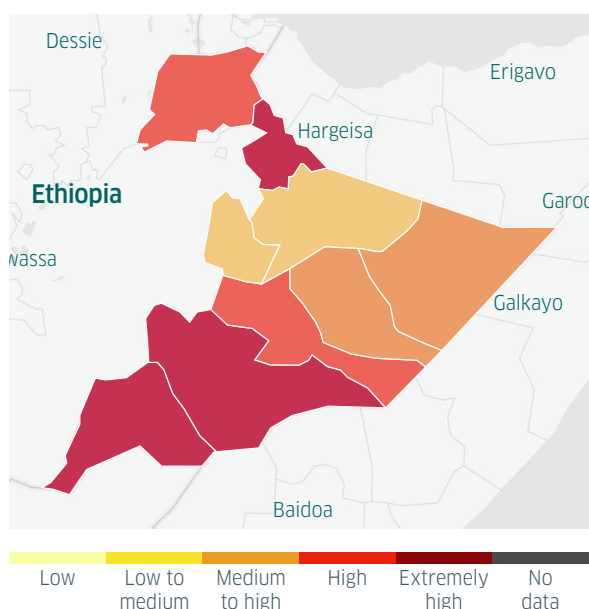
Climate Risk and Vulnerability in Somali and Gambella Regions

Most of the zones in Ethiopia with the highest vulnerability are located in Dire Dawa, Gambella, Somali, Oromia and the former SNNPR regions. Two of these regions, Gambella and Somali, host the largest refugee populations in Ethiopia: 394,202 (Gambella) and 346,007 (Somali). The Somali region has three distinct refugee hosting regions – Jijiga to the east with three long-established camps; Dollo Ado in the south with five camps; and the remote south east of the region where almost 100,000 people crossed from Somalia in 2023.

Somali Region²⁶

Stretching along the southern and eastern border with Somalia, both lower and higher emissions scenarios show that the **Somali region will witness the highest increase in very hot days over the next decades compared to other Ethiopian regions. Extreme precipitation events are expected to become more frequent and more intense.** Although mean annual rainfall may remain stable or even increase over the long-term, the incidence of droughts and flooding is expected to increase throughout the century. With more heatwaves, higher rainfall variability and an increase in heavy precipitation events (HPE), the Somali Region is and will remain in the future highly exposed to the extreme weather events not only of droughts but also extremely high risk from flooding (see Figure 3).

Figure 3: Flood risk – Somali region



The Somali region's climate has already become drier, hotter and more unpredictable over the past years, and is expected to worsen in the future. It is already experiencing the adverse effects of climate-induced droughts, displacement and water stress. When rain came after three consecutive years of historically severe drought in May 2023, then again in October and November of the same year, the Somali region that had suffered from drought – were again hit by 'unprecedented floods.'^{27,28,29} The Somali region was worst affected area for the recent floods, with Somali accounting for more than 80 percent of the affected population. The most substantial impacts in the region have occurred in Shabelle, Afder, Liban, and Dawa zones. Widespread damage to homes, roads, and other infrastructure has been reported in affected areas; twenty-four main roads and bridges have been damaged across the region.

26 L. Destrijcker et al. (2023) Weathering Risk: Climate, Peace and Security Study: Somali Region, Ethiopia. Adelphi Research. https://adelphi.de/system/files/document/climate_peace_security_study_somali_region_ethiopia.pdf

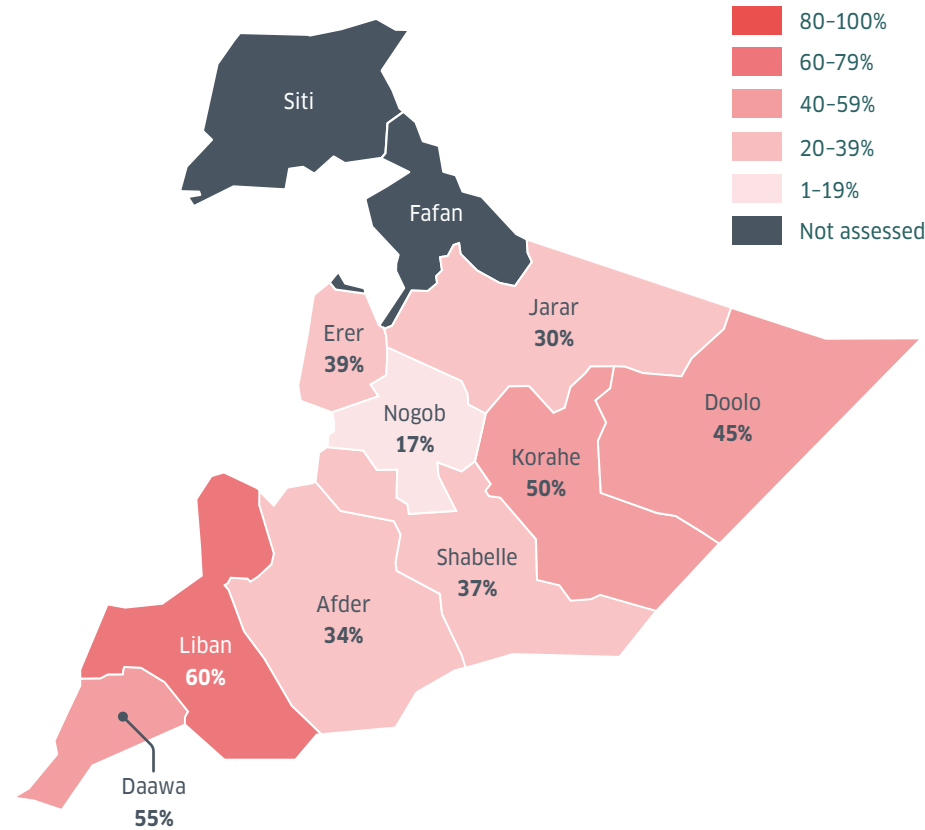
27 T.A.Tufa (March 2024), Conflict, climate change and displacement in the Somali Region of Ethiopia, Humanitarian Practice Network, Issue 84, Article 11. <https://odihpn.org/publication/conflict-climate-change-and-displacement-in-the-somali-region-of-ethiopia/#:~:text=The%20Somali%20National%20Regional%20State%20of%20Ethiopia%20hosts%20102%2C000%20IDPs,displaced%20due%20to%20internal%20conflict.>

28 UNHCR Regional Climate Action Plan (2023-2028) East and Horn of Africa and Great Lakes Region, link

29 L. Destrijcker et al. (2023) Weathering Risk: Climate, Peace and Security Study: Somali Region, Ethiopia. Adelphi Research. https://adelphi.de/system/files/document/climate_peace_security_study_somali_region_ethiopia.pdf

The high dependence on pastoralism and rain-fed agriculture makes the Somali region highly vulnerable to various climate shocks, especially drought. Research conducted in the Somali region in December 2023³⁰ reported a sharp decline in the number of households reporting owning livestock since the beginning of the droughts in 2020, from 43% to only 18% due to the death of livestock during the droughts. In 2023, 62 percent of the population – 3.9 million people – were estimated to need emergency food assistance in the region. The number of people facing critical water shortage and requiring emergency water response was estimated at twenty-nine million (March 2023).³¹

Figure 4: Percentage of households reporting difficulties or shocks in September–December 2024



The Somali region is also facing a major threat to its grazing areas and livestock from the rapid expansion of a pernicious invasive species, *Prosopis Juliflora*, for which there is not effective management (see Box 1). *Prosopis* is climate-resilient, well adapted to the drier and hotter conditions of climate change, withstands floods and grows well in degraded grasslands. It is well suited to the projected climate-related changes that will be occurring and is likely to transform the current landscape into an area that will support far fewer livestock.³²

Figure 5: *Prosopis* in Somali, Ethiopia



30 The research did not include the refugee communities; its focus was on the host community and IDPs.

31 Reach (March 2024) The Impact of Drought and Climate-Related Shocks on Livelihood Practices in Somali (Factsheet) https://repository.impact-initiatives.org/document/reach/0579137a/REACH_ETH_Factsheet_SomaliLivelihoods_1504.pdf

32 <https://thewaterchannel.tv/videos/prosopis-in-somaliland-a-tree-as-threat-and-opportunity/>

Gambella

In the western part of Ethiopia along the border with South Sudan, the Gambella region is one of the most marginalised and conflict-ridden regions in Ethiopia. Clashes occur frequently between the two largest ethnic groups, the Anuak and the Nuer, with tensions exacerbated by the fact that the large refugee population in Gambella are ethnic Nuer from South Sudan.³³ Gambella is also one of the most vulnerable regions to climate change. Between 1971 and 2020, Gambella experienced an average annual precipitation decrease of over 130 mm/decade (about 10%).³⁴ However, climate risk projections suggest the major climate risks will be from **flooding** and for agriculture, **rising temperatures**. The changing rainfall pattern in combination with warming trends could make rainfed agriculture riskier and reduce food production in the region.

The region has experienced erratic rainfall, drought and incidence of floods due to the over-flow of major rivers. The overflow of major rivers and the resultant flooding displaces people, destroys crops and seriously disrupts economic activities. The frequency and magnitude of flooding in the region has increased during the last twenty years.³⁵ Heavy rains between August and October of 2022 in the lowlands in the west of the country caused flooding in Gambella, displacing 185,200 people and affecting an additional 79,631 people through loss of crop land; death of livestock; destruction of property, schools, and health facilities; and contamination of water supply.³⁶

Current climate risks are exacerbating the tensions and conflict in some areas that are the result of historical tensions, the population of the refugees larger than the host communities, scarce resources and what is perceived to be a greater distribution of services to refugees by the host community. These conflicts in turn increase the vulnerability of both hosts and refugee communities to current and future climate risks.

Box 1: *Prosopis juliflora*: A major invasive threat fuelled by climate change

Prosopis juliflora is considered one of the world's most threatening invasive tree species. Originally from South America, it was introduced in Eastern Africa in the late 1960s through the early 1990s to provide wood and fodder and reduce erosion on degraded lands. Unintended and unanticipated consequences have been its rapid spread with a range of severe social and environmental impacts. The rapid spread is largely driven by its prolific seeding (with seeds passing undigested by livestock and other animals), ability to regenerate in high densities and lack of coordinated strategies and actions to manage and control it. Since *Prosopis* cannot compete with good grass cover, overgrazing with increased frequency of drought gives *Prosopis* a competitive advantage.³⁷

Once established, *Prosopis* is well adapted to areas such as Somali with low rainfall and long periods of drought, able to tap groundwater or any other water source and absorb moisture through its foliage during light rains or from dew or with atmospheric sources of moisture. Groundwater abstraction (i.e. the process of extracting ground water) by *Prosopis* affects both surface runoff, groundwater recharge and evaporation losses.³⁸ With no natural predators its aggressive growth creates a challenge for agriculture and pastoralism across Somali and the Afar regions.³⁹ These include the loss of grazing land, loss of access of water, declining biodiversity, reduced human health, an increased human-wildlife and human-human conflicts.⁴⁰

33 S.Z. Hagos (2021) Refugees and local power dynamics: the case of Gambella Region of Ethiopia. Bonn: German Development Institute. <https://doi.org/10.23661/dp25.2021>

34 USAID (2024) Climate Change Risk Profile Ethiopia. <https://www.climate-links.org/resources/climate-risk-profile-ethiopia>

35 Getahun et al. (2021) Determinants of climate variability adaptation strategies: A case of Itang Special District, Gambella Region, Ethiopia. *Climate Services*, Vol.23, August 2021. <https://www.sciencedirect.com/science/article/pii/S2405880721000339#:~:text=The%20Gambella%20Peoples%20National%20Regional,over%2Dflow%20of%20major%20rivers>

36 World Bank Group (2023) Country Climate and Development Report (CCDR). <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/099022224202134449/p17952512670c208619205179e2783cef5>.

37 <https://www.sciencedirect.com/science/article/abs/pii/B978012823320700016X>

38 <https://www.sciencedirect.com/science/article/abs/pii/B978012823320700016X>

39 <https://www.landerprosopis.com/history-of-prosopis-juliflora>

40 <https://www.cabi.org/news-article/three-ways-to-fight-invasive-prosopis-juliflora-tree-in-eastern-africa-all-proved-very-effective-new-study-shows/>

Prosopis has been found to affect animal health in the invaded regions. By replacing grasses, herbs and shrubs, Prosopis harms livestock with its poisoning thorns and causes goat teeth to rot and fall out because the small seeds get stuck between the teeth with the goats dying of starvation following tooth loss.⁴¹

It is estimated that Prosopis currently covers over 6000 sq. kilometres (600,000 ha) of the Somali region, with an annual growth rate of 5 percent.⁴² However, estimates based on research in similar areas in Kenya suggest it may spread at an exponential rate of between 4% to 15% per year.⁴³

The spread of Prosopis in Ethiopia will quicken with climate change. Under the current climate, 94.8% of Ethiopia is non-suitable for Prosopis while only 0.4% (4.56 million ha) is highly suitable. In 2050, highly suitable is expected to increase by 55.6% and 63.66%, while moderately suitable area is projected to increase by 33.3% and 42.9%.⁴⁴ Compared to the current climatic condition, in 2070, highly suitable areas is projected to increase by 73.3% (3.43 million ha) and 80.0% (3.65 million ha). This further expansion will lead to greater environmental and economic damage, further threatening the livelihood of pastoral communities and their livelihoods.⁴⁵

Prosopis has been found to affect animal health in the invaded regions. By replacing grasses, herbs and shrubs, Prosopis harms livestock with its poisoning thorns and causes goat teeth to rot and fall out because the small seeds get stuck between the teeth with the goats dying of starvation following tooth loss.⁴⁶

Vulnerability of Refugees

Refugees are experiencing not only the bio-physical-economic risks associated with climate change in their areas, but also the socio-economic constraints of their refugee status. As is found world-wide, refugees in Ethiopia are poorer, marginalised and politically disenfranchised when compared to the host population. Despite impressive reforms and pledges made by the Government of Ethiopia, refugees still lack basic socio-economic rights. Refugees in Ethiopia are now allowed to have SIM cards and to open bank accounts, but reforms introduced to give refugees the right to live and work in Ethiopia, and to run legal, registered businesses, have not yet been fully implemented. In practice most refugees still face steep social and economic hurdles due to their refugee status – restrictions on movement and their ability to work; unable to formalise businesses; no access to bank loans for small businesses. Ethiopia's policy direction is towards more socio-economic inclusion of refugees into Ethiopian society, but operationalisation of the more inclusionary policies is lagging, hindering the pace of the anticipated inclusion. This situation faced by the majority of refugees places them at greater climate risk. Lack of basic socio-economic rights translates into less resilience to severe weather events that are undermining agriculture, pastoral livelihoods and food security. Building the adaptive capacity of refugee and host communities is crucial to withstand climate variability and change.

Humanitarian Assistance

Humanitarian assistance to refugees living in a situation of protracted displacement in Ethiopia has been steadily dwindling. The number of forcibly displacement people living in Ethiopia, however, has increased. Available humanitarian budgets primarily support short-term emergency interventions which are commonly perceived as being core to their purpose – food aid, basic services such as health, water and education, protection and nutritional support. However, budgets are insufficient and institutions are not

41 W. Shiferaw and S.Demissew (2022) Effects of Invasive Alien Prosopis juliflora (S.W.) DC and its Management Options in Ethiopia: A review. <https://www.intechopen.com/chapters/85085>

42 <https://www.landerprosopis.com/history-of-prosopis-juliflora>. In the Afar region Prosopis has invaded 1.2 million hectares of land since its introduction in 1990

43 <https://www.sciencedirect.com/topics/agricultural-and-biological-sciences/prosopis-juliflora>

44 The difference is the result of the projection of two climate scenarios – RCP4.5 and RCP8.5. Given current emissions, the more moderate RCP4.5 is unlikely to occur.

45 Sintayehu, D.W., Dalle, G., and Bobasa, A.G. (2020) Impacts of climate change on current and future invasion of prosopis juliflora in Ethiopia: environmental and socio-economic implications. *Heliyon* 6 (8), e04596. [https://www.cell.com/heliyon/fulltext/S2405-8440\(20\)31440-7?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2405844020314407%3Fshowall%3Dtrue](https://www.cell.com/heliyon/fulltext/S2405-8440(20)31440-7?_returnURL=https%3A%2F%2Flinkinghub.elsevier.com%2Fretrieve%2Fpii%2FS2405844020314407%3Fshowall%3Dtrue)

46 W. Shiferaw and S.Demissew (2022) Effects of Invasive Alien Prosopis juliflora (S.W.) DC and its Management Options in Ethiopia: A review. <https://www.intechopen.com/chapters/85085>

structured to support the integration of humanitarian assistance with longer term strategies supporting local development – access to clean energy, sustainable water supply systems, and education and health services.⁴⁷

It is increasingly recognised that the classic humanitarian response model cannot sustain basic services to refugees in a protracted situation and that more systemic and sustainable approaches are needed. The humanitarian system is also not fit-for-purpose to deliver sustainable economic interventions, especially when considering the high climate risk faced by both refugee and host communities. Humanitarian strategies and programmes focus on protection and the provision of life-saving assistance, with until recently little attention given to climate risk (except within the scenario of climate refugees) and sustainable adaptation. While this is changing, there are missed opportunities for the climate-proofing of infrastructure and the implementation of climate resilient sustainable activities in the refugee setting. Humanitarian agencies are turning to other international agencies and organisations to provide the expertise, skills, and financial resources that are needed for sustainable development and building resilience to a rapidly changing climate.

The humanitarian refugee response in Ethiopia is attempting to address the transition to long-term rather than emergency programmes, moving from a camp based approach to one focused on settlements and integration. However it continues to be primarily humanitarian in nature with limited developmental approaches to catalyse the private sector for a more sustainable response that goes beyond care and maintenance and promotes self-reliance. This is where market systems development programmes such as SHARPE plays a role, leveraging the private sector for growth and development in refugee hosting areas in Ethiopia.



47 UNHCR (2024) Melkadida Refugee Compact. <https://data.unhcr.org/en/documents/details/107301>

Strengthening Host and Refugee Populations in Ethiopia – SHARPE

SHARPE is a market systems development programme, funded by the UK Government Foreign, Commonwealth and Development Office (FCDO), designed to create the conditions for greater economic inclusion of both refugees and remote, rural host communities in the wider economy. SHARPE is working with refugee and host communities in some of the most climate vulnerable regions of Ethiopia – in the refugee hosting woredas in Gambella, and Jijiga and Dollo Ado in the Somali region. SHARPE partners with the private sector, co-investing in market expansion and business growth into refugee hosting regions of Ethiopia.

More inclusive refugee policies and provisions introduced by the Government of Ethiopia, as well as pledges made at successive Global Refugee Forums, have provided the scope and space for programmes such as SHARPE to promote inclusive and sustainable economic development for refugee and host communities. Using an adapted Market Systems Development (MSD) approach, SHARPE has been working to facilitate private sector investment and market linkages into remote refugee hosting regions to generate inclusive economic opportunities for both refugees and host communities. SHARPE has catalysed investment in strategic engagement areas in agri-business, solar energy, and finance. SHARPE offers co-investment, technical assistance and capacity development to partners on a cost-sharing basis.

SHARPE's adapted MSD approach has generated notable successes with sustainable commercial businesses developed among both host and refugee communities in egg and poultry production, livestock (goats), agro-vets and agri-inputs, digital finance and solar energy. These micro-businesses operating inside and around the camps are linked with regional MSMEs to access a range of support services, including finance, technical support and essential inputs, creating market relationships and supply chains that are sustainable beyond the lifetime of the SHARPE programme.

Climate Risk Assessment

Overview - Climate Risk, Adaptation and Resilience. This climate risk assessment of the SHARPE programme focuses on the climate risks associated with the regions in which it is working and its programme, the contribution of its current programme to climate resilience, adaptation, and mitigation, and recommendations on how to further strengthen resilience and enable local adaptation to climate risks and mitigation (lower emissions). Underpinning the recommendations is the understanding that while climate change will bring more frequent and intense extreme weather events such as heatwaves, droughts and floods, households and communities can reduce their vulnerability and increase their resilience by taking action to prepare for and adjust to the current and projected impacts of climate change.

Adaptation in this context refers to the capacity of communities and households to adjust their economic system in response to actual and/or expected climate changes/extremes in order to moderate potential damage and/or to benefit from opportunities associated with climate change. Adaptation activities provide the resources, tools or skills to better cope with changing, unpredictable and extreme weather. While adaptation is the process of adjusting to the actual or expected climate and its effects, resilience is the actual capacity to prepare for these effects, particularly the capacity to 'bounce back' from destructive climate-related events⁴⁸.

An important adaptation strategy that individuals and households often employ to manage the effects of climate shocks on their way of life is livelihood diversification, in which households move away from relying on a single source of income and engage in a range of activities to ensure their survival and improve their living standards. Livelihood diversification includes income-generating activities on-farm (e.g. growing crops, livestock, hunting and fishing) and off-farm, such as agroprocessing, trade, wage labour and remittances.

48 https://www.ipcc.ch/site/assets/uploads/2018/02/WGIIAR5-Chap20_FINAL.pdf

An important role that programmes such as SHARPE can play to facilitate climate adaptation and strengthen resilience is to support livelihood diversification by resolving the financial bottlenecks experienced by households, and initiating the establishment and expansion of micro and small enterprises.⁴⁹

Climate risk. SHARPE works in the Gambella and Somali regions which are two of the most vulnerable regions in Ethiopia. The livelihoods of the refugee and host communities and the current and anticipated climate risks significantly differ in these regions.

Low and humid, encompassing several river basins and the country's sole tropical mountain rainforest, the greatest climate risks in Gambella will be from **erratic rainfall and floods**, such as the recent floods in 2023, which will affect agriculture and livestock. For the host community, livestock and crop production are the main means of livelihoods for the majority of the residents and to lesser extent fishing and bee keeping. Flood recession agriculture is common, particularly maize and sorghum, being widely practiced along the rivers. Livelihood activities differs based on ethnic group (e.g. the Nuer practise agro-pastoralism; the Anuak are predominantly sedentary agriculturalists) and geographic area. The refugee population is primarily Nuer with people practising a range of small-scale informal livelihood strategies to buy food and basic necessities – backyard farming, including small-scale goat and poultry farming; day labour; petty trade; selling firewood and charcoal; brewing beer etc.

In the arid **Somali** region, refugees primarily reside in one of two zones, the eastern Fafan zone near the regional capital, Jijiga, and the southern Liban zone, close to the Somali and Kenyan borders, and including the border town of Dollo Ado. Dollo Ado has been referred to 'as one of the epicentres of the climate crisis' (see Box 2). Climate projections for the Somali region suggest a **drier, hotter and more unpredictable climate**. Within the Somali region livelihoods of the host communities are mainly pastoralism and agro-pastoralism, both of which are vulnerable to the primary climate risks of drought and rising temperatures. Refugee livelihoods are constrained by the lack of access to land for agriculture, low levels of assets and livestock ownership as well as no access to finance. They do, however, live in close proximity to international borders with strong cross-border clan and family connections – there is a thriving cross-border economy, particularly in Dollo Ado, with many refugees using their connections in Somalia to bring goods back into Ethiopia for onward sale. While Jijiga refugees live in relatively close proximity to the regional capital and economic centre of the Somali region, those in Dollo Ado have access to the waters of the nearby Genale river for irrigation and for fishing, although proximity to the river also brings a flood risk.

Box 2: Dollo Ado⁵⁰

UNHCR refers to Dollo Ado, as one of the epicentres of the climate crisis ravaging the Horn of Africa, noting that after five consecutive failed rainy seasons, the most recent drought was the worst to hit the region since 1981. This resulted in water shortages that destroyed thousands of acres of crops, killed hundreds of thousands of livestock, and dried up water sources in many communities. The drought was followed by dangerous flash floods and the destruction of agricultural fields, and infrastructure – water systems, irrigation, roads and bridges.

49 https://www.sipri.org/sites/default/files/2024-04/pb_2404_mistra_livelihood_diversification_v2.pdf

50 UNHCR (2024) Melkadida Refugee Compact. <https://data.unhcr.org/en/documents/details/107301>

Climate Risk Assessment of SHARPE Interventions

SHARPE has worked to build more inclusive markets and business opportunities for hosts and refugees in the following sectors: agri-business (poultry; livestock; crops; agro-vet services; feed); finance; energy. This assessment focuses on the interventions that were implemented in Dollo Ado – small-scale poultry production; sheep and goat trading/fattening; agro-vet services; solar energy; and financial inclusion. Interventions on climate smart agriculture, implemented together with Shayashone plc in Gambella, were not included in this assessment.

Climate change poses a unique set of risks for SHARPE's agri-business sectors, particularly for poultry and livestock. Changing temperatures, rainfall patterns and the availability of food and water make keeping an animal more difficult and uncertain. Increasing temperature is likely the greatest climate change factor affecting poultry. Poultry are particularly vulnerable to climate change because birds can only tolerate narrow temperature ranges with poultry production decreasing at temperatures higher than 30°C. Heat stress on poultry reduces feed intake, weight gain, carcass weight, and protein/muscle calorie content. Laying hens respond to increasing temperatures by reducing the number and size of eggs produced. Stressful weather conditions negatively affect internal and external quality of eggs. High temperatures result in lower fertility rates, egg hatchability, and chick quality. Additionally, heat-stressed males produce semen with lower sperm quality and concentration. To adapt to higher temperatures, potential adaptation actions include the introduction of more heat tolerant poultry breeds and construction of poultry housing systems (e.g. poultry sheds) that reduce the risk of heat stress.⁵¹

For pastoralists and agropastoralists in the Somali region, climate change is a pressing threat and one that challenges their immediate and long-term livelihoods – increased temperature and lower/ erratic rainfall patterns are associated with reduced feed resources, greater animal mortality rates, smaller herd sizes, fewer water sources during the dry season, decreased animal performance and productivity, and emergence of new animal diseases. In Gambella the changes in rainfall patterns, flooding and rising temperatures pose climate risks associated with floods and wet feed and fodder but with similar results as in Somali of greater animal mortality, decreased animal performance, and increased presence of diseases and pests. Often cited adaptation actions for livestock include shift in livestock species breeds that are resistant to drought, heat and harsh environments (e.g. small ruminants,⁵² poultry), disease control (e.g. improved access to veterinary services), and cooling (e.g. cooler goat sheds).⁵³

Poultry

SHARPE is supporting the growth of commercial poultry farming in refugee and host communities in both Gambella and Somali regions. SHARPE's work on poultry farming builds upon pre-existing practices and skills within both host and refugee communities, allowing people with some experience of small-scale backyard poultry farming with local poultry breeds, to upgrade to commercial poultry farming. Keeping a small number of local chickens for eggs is common practice among all the refugee camps in Gambella and Somali regions. The chickens are mainly managed by women and the business is small-scale and informal, with most refugees only able to afford to keep a small number of live chickens.

Through support from the SHARPE programme more than 50 small-scale poultry farmers, both hosts and refugees, have scaled up their business and established commercial poultry farms in their compounds. This includes an investment in improved breeds and in well-ventilated poultry sheds, a marked improvement on traditional poultry sheds. These poultry farms are linked to, and supplied by, local poultry firms as well as to other local businesses which provide a range of support services such as feed, medicines and finance on a commercial basis. In Gambella commercial poultry farms are mainly producing live pullets for sale and in the Somali region the main business is egg production.

Poultry farms established and run by refugees are all small enough to be accommodated within the owners' compound inside the camp, supplying eggs to the local markets and for domestic use.

51 <https://www.thepoultrysite.com/articles/climate-change-and-poultry-production>

52 When compared to cattle which need more water and access to pasture, small ruminants such as goats and sheep are better adapted to survive and reproduce in regions experiencing the hotter and drier effects of climate change.

53 https://unfccc.int/sites/default/files/resource/2_FAO_Carolyn_UNFCCC%20Koronivia%20Nov%202020_Livestock%20Adaptation.pdf

SHARPE's Intervention Design Analysis. The initial sector analysis which led to the design of the poultry intervention identified the key challenges encountered by commercial poultry production in these regions as lack of capital, lack of knowledge to manage a commercial poultry farm and lack of access to quality inputs – day-old chicks, feed, vaccines, medicines and extension/advisory services.

To address these constraints:

- SHARPE identifies hosts and refugees with relevant skills and interests to establish a commercial poultry farm. SHARPE provides grant finance to co-finance the start-up and initial running costs. The support includes finance for shed construction and to procure inputs from a local poultry firm, or regional poultry hub (below).
- The regional poultry hub (a regional/local poultry company) provides:
 - Training on how to manage the technical and commercial aspects of their business.
 - Training on how to construct their poultry shed – sheds are essential for bringing up healthy stock.
 - Inputs to each producer on the completion of their shed: chickens, grower feed, feeder and 'drinkers' (water tanks).
 - Preferred chicken varieties. Appropriate poultry varieties were selected to meet the different market preferences, eggs in Somali region and live chickens for meat in Gambella

Key climate risks to poultry production and how to tackle the risks

Climate risk. In the Somali region increasing temperature is the greatest climate risk factor affecting poultry. Poultry are particularly vulnerable to climate change because birds can only tolerate narrow temperature ranges with poultry production decreasing at temperatures higher than 30°C. Heat stress on poultry reduces feed intake, weight gain, carcass weight, and protein/muscle calorie content. Laying hens respond to increasing temperatures by reducing the number and size of eggs produced. Stressful weather conditions negatively affect internal and external quality of eggs. High temperatures result in lower fertility rates, egg hatchability, and chick quality. Additionally, heat-stressed males produce semen with lower sperm quality and concentration. Adaptation actions to heat stress include the introduction of more heat tolerant poultry breeds and construction of poultry housing (e.g. poultry sheds) that reduce the risk of heat stress, slower egg production, diminished egg quality (e.g. thin shell) and reduced weight gain of chickens produced for meat.

Adapting to rising temperatures and heat stress: climate-proofing poultry sheds. Reducing the heat stress of poultry is central to effective adaptation. The current poultry sheds are well constructed and well-ventilated (walls are half screen) with 'drinkers' and water tanks (50 Litre) to provide water. They currently serve their intended purpose which is to provide a safe, comfortable and healthy environment for the chickens.



However, in places such as Dollo Ado where the climate is currently hot and dry, and climate projections are for it becoming both hotter and drier, these sheds will not be able to provide the cooler temperatures and water that will be needed to lower the heat stress of poultry. To become climate proofed for rising temperatures, the sheds should be redesigned to include insulation of the roof, fans, a larger water tank and a more efficient water-delivery system that will provide water throughout the day. In hotter conditions poultry sheds will also need a built floor (rather than earth floors used for laying) for ventilation during extended period of high temperature.

Practical implications of adapting to climate risk for poultry farming: lower heat stress but higher infrastructure costs. Future climate considerations should be taken into account in the design of infrastructure such as poultry sheds. The 'shed package' offered by SHARPE should be modified to protect the chickens from high temperatures and make the sheds 'climate proof' – provide more ventilation, including a fan, as well as protection from the sun and improved water efficiency. Ideally the sheds should include household solar panels to provide the power to operate the fans, particularly in off-grid situations.⁵⁴

Climate driven modification of poultry sheds will significantly increase their costs. Most refugee camps are currently off-grid and would require solar panels to power a fan. Modern poultry sheds suitable for commercial level farming are already expensive and a significant entry barrier to cash-poor households such as refugees. Climate proofing poultry sheds in small-scale farms operated by refugees would result in an approximate 50% increase in construction costs. These climate proofed sheds are unlikely to be affordable for either refugees or host communities without external grant assistance.

Sheep and Goat Trading

Sheep and goats play an important role in household livelihoods among both hosts and refugees. It is common practice among both refugee and host community households to invest in a small number of goats not only for production, but as a means for saving income (e.g. 'savings on the hoof'). This is a very common livelihood and resilience strategy pursued by many households across both refugee and host communities in both Gambella and Somali regions. Many households keep a small number of small ruminants with fattening for onward sale a popular livelihood activity, largely pursued by women, across both regions.

Building upon this common livelihood strategy SHARPE has strengthened the local markets in both Somali and Gambella regions for sheep and goat trading. The purpose of the intervention is to benefit refugee and host communities who keep small numbers of sheep and goats by boosting income through increased sales of healthier livestock. It is intended to allow people with the right skills and experience in the buying/ selling and management of sheep and goats to upscale their small businesses to a more commercial level and engage in wider market systems. SHARPE has also worked to build the market systems that are necessary to support sheep and goat trading – local production of nutritious feed and fodder; access to finance; and access to veterinary supplies such as medicines and vitamins; and peripartetic veterinary services.



54 Sheds will be 'off-grid' or if on a UNHCR mini-grid will only receive electricity a few hours a day. The training and installation of the solar panels can be facilitated by SHARPE's solar energy intervention.

SHARPE's Intervention Design Analysis: The intervention design analysis identified the key challenges as a lack of capital for business growth, refugees being excluded from the markets due to the small size of their businesses/small number of animals, low availability of medicines and vaccines to manage the health of the animals, and knowledge of livestock nutrition, rearing and best practices.

To address these constraints:

- SHARPE offers micro-grants to hosts and refugees to scale up their sheep/goat business – to construct an improved shed; to procure animals to boost the stock; and for training in the health and nutritional for fattening of sheep and goats
- Traders construct improved sheds for livestock fattening and trading.
- SHARPE designed and implemented an intervention focusing on increasing availability of medicines and vaccines through improvement and extension of agro-vet services (see agro-vets below)
- SHARPE also established local manufacturers of quality animal feed in the refugee hosting regions, in partnership with EthioFeed.

Key climate risks to livestock production and how to tackle the risks

The main climate risks are from more frequent (and severe) weather extremes, floods and increased rainfall/wetter conditions. Goats are vulnerable to diseases associated with wet hooves and may develop mouth infections from wet fodder (if wet fodder is cut and carried to the shed).

Adapting to floods and wetter conditions: climate-proofing goat sheds. In regions such as Gambella which are predicted to become wetter and increasingly vulnerable to floods care needs to be taken to protect infrastructure, such as goat sheds which provide shade and water, from floods and rainfall. As an adaptation action, shed floors should be raised to protect the goats from flood waters and wet floors. Fodder storage facilities should also be constructed with floods and heavy rainfall in mind. Small solar energy generated driers for fodder could be included in the shed package for co-financing. Other adaptation actions are to promote trade and market access, and improving animal health through access to veterinary services and medicine.

Practical Implications of adapting to climate risk for sheep and goat trading/fattening

As with poultry farming, any infrastructure constructed for the housing and management of sheep and goats needs to consider current climate trends and future climate scenarios, in this case wetter conditions caused by floods and increased rainfall. Specifically sheds constructed for sheep and goats should include a raised floor and fodder needs to be kept dry. Traditional practices of keeping goats (e.g. tethering) will also not protect the animals from floods and wetter conditions. This will, however, entail an additional cost to build a quality shed that can withstand flooding and protect the animals from excess water – the cost will increase by approximately 45-50%, making these unaffordable for most refugees and poor host community households without external financing.

Agro-vets/Agri & Veterinary Inputs

To provide a sustainable supply of farm inputs, livestock supplies (vitamins, medicines etc) and veterinary services SHARPE has worked with local agri-input companies and local agro-vet dealers, improving the goods and services they offer and extending their reach into refugee camps. In the case of Dollo Ado the agro-vets are mainly refugees, operating small shops inside each of the five camps, providing services to both hosts and refugees. Refugees have also been trained and certified as Community Animal Health Workers (CAHWs), working with the agro-vets and providing community based animal health services for a small fee.

This intervention has made quality agricultural and veterinary inputs available and accessible to refugees which in turn allows them to invest in their farm and livestock-based businesses, boosting animal health and farm productivity.

SHARPE's Intervention Design Analysis. For veterinary inputs, the intervention design analysis identified the key challenges as availability of veterinary drugs and feed, access to agro-vet services for host communities and refugees outside the main towns, and limited knowledge and skills on livestock drugs use (retailers have little knowledge of what they are selling due to the disconnect from suppliers).

To address these constraints:

- SHARPE offered finance to allow agro-vet dealers to invest in and expand their shops, improve the goods and services they can offer and grow their customer base in both refugee and neighbouring host communities. SHARPE also financed and supported two agro-vets per region to become regional wholesalers, allowing them to supply smaller agro-vets in the region.
- SHARPE provided training for both hosts and refugees to become certified community animal health workers (CAHWs). This provides jobs to the CAHWs who work closely with the agro-vets as sales agents, providing veterinary and animal health outreach services within their communities.
- Linked agro-vets with agri-inputs importers and suppliers in Addis Ababa to establish new product supply lines. Agro-vets were also linked with financial institutions for business loans, allowing them to develop a credit history and track record with the financial institutions.

Key climate risks to agro-vet and animal health goods and services

Drought and generally drier conditions pose the main climate risks to businesses offering goods and services for livestock. As was witnessed during the recent extended drought in the Somali region, people moved their herds to different regions, in many cases moving to peri-urban and urban centres (Dollo Ado, Melkadida, Jijiga) in search for support services. In some cases pastoralist families decided to transition to a more settled life in peri-urban centres, effectively giving up on the traditional pastoralist way of life. Drought resulted in the death of many animals and there was a drop in livestock and incomes. All this results in a decreased demand for agro-vet goods and services. Agro-vets and community animal health workers, were, however a critical part of the support system to try and keep animals healthy and alive during the drought. By staying open and continuing to offer their services throughout a period of drought they contributed to the climate resilience of households who are dependent upon livestock for their livelihoods.

Practical Implications of climate risk for agro-vets and community animal health workers (CAHWs)

Agro-vets and community animal health workers are dependent on customers being able to afford their goods and services. For people such as refugees who are living in remote, marginal regions climate shocks are likely to increase their need for agro-vet services due to the impact of heat and water stress and the increase in incidence of animal disease and parasites and pests (e.g. ticks). Conversely, however, the climate risk may reduce their ability to pay for the services and lead to a reduction on what the agro-vets and community animal health workers are able to provide, particularly to more remote areas.

During extreme climate events such as extended droughts commercial suppliers of agricultural and veterinary outreach services (agro-vets and CAHWs) should be considered an essential part of the response. Supporting these businesses to remain functional and continue to offer their services during and beyond a crisis will contribute in the long-term to overall system and household resilience.



Energy

The refugee hosting regions, and camps in both Somali and Gambella are largely off-grid.⁵⁵ With the exception of Kebribeyah camp (where camp residents 'buy' electricity from households in the neighbouring host community) all refugee camps are off-grid. For the majority, people are forced to rely on batteries and/or buying electricity from people running diesel generators. Both of these options are expensive and polluting, harmful to the environment and contributing to GHG emissions. Owning and running a diesel generator, and selling electricity, is seen as a viable and profitable business for refugees who have enough capital to purchase a generator.

To provide access to clean energy SHARPE is implementing two interventions on solar energy: solar kits and energy kiosks.

Solar kits (household and productive use). SHARPE is working with two solar energy companies to assemble and sell quality solar energy kits to refugees and host families in the Somali region at commercial rates. These kits are designed for home usage, powering three bulbs, a torch and a phone charging station; or for productive use to be used by small businesses to provide phone charging services and/or run a small barber shop. All are sold using the PAY-Go model of monthly instalments paid through digital finance, with a reduced down-payment for refugees to make them affordable to people with minimal monthly incomes.

The purpose of this intervention is to benefit refugee and host households by facilitating access to solar energy (kits), which in the medium term is cheaper and cleaner than buying from a diesel generator or buying batteries and can provide a new/improved business opportunity by providing electricity services within an off-grid community.

Energy Kiosks. Solar energy kiosks are stand alone, multi-purpose booths, manufactured from fibre glass and powered by solar panels on the roof. They are intended to provide energy services to the local community – phone charging, internet, refrigeration – while also providing a space for the owner to run a small shop/kiosk. SHARPE has worked with a private company to manufacture and supply 15 solar energy kiosks to households in both host and refugee communities in Somali and Gambella regions. Together with the owners of the kiosks (host and refugee households) SHARPE co-funded the purchase of the solar kits – SHARPE paid up-front with the owner/operator paying the rest of the sales cost through a series of PAY-Go payments. The monthly repayment amount was calculated to be affordable from the revenue generated from the kiosks.



55 Ethiopia has the third largest energy access deficit in Sub-Saharan Africa with about half the population still without access to reliable electricity. The grid network covers nearly 60% of town and villages, but only 10% of Somali is on the national grid. Ethiopian electricity production heavily relies on hydropower, which is vulnerable to climate related changes to hydrological conditions, stream flow, water availability, and reservoir storage. <https://www.worldbank.org/en/news/press-release/2024/04/03/energizing-afe-ethiopia-new-world-bank-program-expands-access-to-electricity#:~:text=Ethiopia%20has%20the%20third%20largest,without%20access%20to%20reliable%20electricity>

SHARPE's Intervention Design Analysis. Although demand is huge for solar products in these off-grid situations, the intervention design analysis identified the key challenges as lack of supply and affordability of quality solar supplies. Low quality, cheap solar supplies are available, often from across the border, with no after sales service but they usually break down quickly. Prior to SHARPE there were no quality home solar kits available for refugees and host community households to buy using a PAY-GO system through digital finance.

To address these constraints:

- SHARPE partnered with three private solar companies to support their market expansion into the refugee markets of Jijiga, Dollo Ado and Gambella.
- SHARPE co-funded the import, assembly and manufacture of solar energy kits and energy kiosks. The kits are intended both for home use as well as for productive use and for sale into host and refugee communities in the Somali region. The kiosks are intended to be run as stand-alone businesses offering energy services commercially.
- Partner companies conduct marketing campaigns and identify, recruit and train sales agents locally, including refugee sales agents. They distribute the kits to the local sales agents who in turn sell the kits to refugee and host community households, earning a commission. They also provide ongoing servicing and maintenance for the energy kiosks.
- People within the host and refugee communities purchase the solar products using the PAY-GO model, with reduced down-payment requirements for refugees.

Key climate risks to solar energy products

Solar products are overwhelmingly positive in terms of environmental and climate impact, providing much needed clean electricity to people living in off-grid situations. Without solar power, people use polluting and more expensive sources for their power – running diesel generators (or buying electricity from someone running a diesel generator) and/or batteries.

In all future climate risk scenarios – hotter, drier, wetter, more unstable – the main risk to solar energy panels is from extreme weather/storms which can damage the solar panels. There may also be additional risks posed by higher temperatures. This is not a major risk and can be addressed simply by providing information at the point of purchase to solar kit owners on how to avoid damage from storms and high temperature. Specific to energy kiosks care should be taken not to locate them in flood prone places.

Box 3: Uba Abduselam

Uba Abduselam Warsame (35) is a Somali refugee, living in Melkadida refugee camp since 2019. She is a single mother of three children aged between 14-17.

In December 2022, SHARPE supported Uba to begin operating an energy kiosk business. She provides a mobile phone charging service, charging customers 10 ETB for a small phone and 20 ETB for a large phone. On average she charges 20 mobile phone per day. In the kiosk, Uba also sells soft drinks and items such as shampoo, body treatments and hair oil. Almost all her customers are refugees. During Ramadan, the ration cuts and when the kiosk is not fully operational (due to technical issues), Uba earns on average 500 ETB per day from her kiosk. This amount increases to 750-800 ETB when the kiosk is running properly.

The energy kiosk business has faced challenges due to technical problems relating to her solar battery which has impacted the capacity of the kiosk. Due to limited solar capacity of the kiosk. Due to limited solar capacity, she is unable to use the refrigerator in her kiosk.

This has meant she has been unable to keep ice or cold drinks. It has had a negative effect on her business especially as she has less business for tea and vegetable sales (her other income generating activities).

Overall the kiosk has enabled Uba to have a higher and more secure income source. Previous to SHARPE intervention Uba was trading on a much smaller scale. The kiosk has enabled Uba to expand her income sources and therefore increase her resilience to shocks. However, solar energy products require maintenance – needing both technical expertise and parts to be available. The remote location of Dollo Ado has meant it is often challenging to find a solution and in addition, Uba has limited resources to afford expensive technical solutions. For Solar energy products to be effective in regions such as Dollo, they need to be supplemented by local expertise to ensure the products are fully utilised. This needs to be considered (and costed) from the early stages of solar-energy interventions.



Financial Inclusion

Digital financial services have only recently become available to refugees – recent refugee reforms allow refugees to own a SIM card and to open a bank account for savings. Digital financial services are much more developed in the Somali region than elsewhere in the country, with the use of mobile money a common practice and multiple (digital) financial service providers competing with one another. SHARPE partnered with one financial institution, Shabelle Bank, to expand their DFS platform, Hello Cash, to the more remote host and refugee markets of Dollo Ado and Jijiga. It is relatively straightforward for refugees to sign up to a DFS service provider and to use that platform to conduct digital transactions (cash-in, cash-out, payments, transfers). Other financial services, however, such as SME financing are still not available for refugees and extremely difficult to access when it comes to rural host communities

SHARPE's Intervention Design Analysis. The interventions design analysis identified the key challenges to business financing and digital financial service as the limited number of agents serving remote host and refugee population, low financial literacy and limited ownership of SIM cards and phones (particularly among women), lack of collateral for business loans, and refugees unable to access business loans and viewed as high-risk clients. Refugees are also unable to formally register their businesses which automatically excludes them from business finance.

To address these constraints:

- SHARPE increased the number of DFS agents operating in remote locations, including inside refugee camps (identification, recruitment and training) as well as increasing the number of local merchants who receive payments digitally for their goods and services.
- Promoted Hello Cash and onboarded more than 70,000 new customers to Hello Cash.
- Provided small business loans to Hello Cash agents, including refugees, with personal guarantors used for collateral rather than fixed assets such as a house.
- Opening four sub-bank branches in/beside refugee camps to increase access to financial services.
- Linked many micro refugee businesses to Shabelle Bank and helped to facilitate business loans to these businesses.

Key climate risks to digital finance services. There are no direct climate change risks to this intervention.

Adaptation. Having access to reliable finance can help to strengthen climate resilience by facilitating the establishment and operations of local enterprises and livelihood diversification.

Climate Risk Summary for SHARPE

SHARPE's work on agri-business is at greatest risk from climate change, particularly poultry and goat interventions as the climate becomes drier, hotter, wetter and more unpredictable. However, poultry and livestock farmers can reduce their vulnerability and increase their resilience to current and future climate scenarios by practical adaptations to farm infrastructure (poultry sheds; goat sheds). By providing shelter that enables increased production and income, poultry and goat sheds constructed with SHARPE support currently contribute to climate adaptation and strengthening climate resilience. However, with relatively small but essential climate-sensitive changes incorporated into shed design and construction, the infrastructure will become better able to meet current and future climate risks and contribute to climate resilience. Additional financing – to make these sheds climate proof and provide solar power – can further lower climate risk and contribute to the sustainable adaptation of the interventions to the changing climate.

As outlined above, the proposed changes to the SHARPE poultry and goat interventions will increase the costs of starting and running a business. This will likely create a barrier to refugee and host business owners who intend to start or expand their businesses. If unable to afford climate proofing measures businesses will be less able to withstand climate shocks. This is particularly challenging when refugee business owners are not able to access finance products such as loans or insurance and are therefore especially vulnerable to shocks.

Contribution to climate adaptation and mitigation

Although climate adaptation and mitigation are not explicitly included in its objectives, SHARPE's interventions are making significant contributions in strengthening the climate adaptation of refugees and host communities and lowering GHG emissions (mitigation). SHARPE is strengthening climate adaptation by increasing income, assets, and diversifying livelihoods, all of which increases the ability to adapt to climate change stresses shocks and variability and/or help reduce exposure to them.

SHARPE is achieving this by providing:

- New opportunities for income that are well suited for refugees – poultry, goat and energy interventions are purposely designed for small compounds and for women's participation. Upgrading infrastructure used in these agri-businesses (i.e. sheds) already provides a higher level of protection for the animals against climate extremes than would otherwise be the case. Financial inclusion - access to finance is vital to community resilience, enabling vulnerable populations to prepare for severe climate-related events and recover from them. Digital financial services contributes to adaptation by increasing accessibility, enhancing efficiency and the speed of transaction, and enabling the financial inclusion of refugees. It provides income and supports the development, distribution or adoption of new technologies that will enable access to finance. Business financing for agri-business plays a significant role in their success and are essential for providing the resources that enable sustainable adaptation
- New 'off-grid' solar systems that allows delivery of renewable energy directly to houses, businesses and/or community services and diminishing use of diesel generators – contributes not only to adaptation but also mitigation.

Livelihood diversification with multiple sources of income is at the centre of adaptation, enabling households to spread risk and being able to piece together a livelihood from different income streams. Even if one activity brings in more income, others may often be initiated or continue to be pursued. Diversification is a traditional adaptation to the high dependency on rainfall patterns. In the Somali region pastoralist, agro-pastoralist and crop producing communities diversify their income and food sources to sustain their livelihoods (see Box 4 below). Households today rely not only on livestock and crop farming but also on remittances and gifts, casual labour, own business(es), salaried work loans or humanitarian aid as other primary income or food source.⁵⁶

Box 4: Case study – Diversification > Adaptation

Abubakar Hassen, is a 45-year-old Somali refugee living in Buramino Camp, Dollo Ado. He arrived in Ethiopia in 2009 when he was forced to leave due to conflict and drought in his homeland. Before migrating, Abubakar owned multiple business ventures including in agriculture, livestock and retail.



As a refugee entrepreneur Abubakar manages several businesses which provide him with different income sources. In 2022, SHARPE supported Abubakar with a 250,000 ETB grant (£3,900) for shed construction and purchase of 20 goats. Through this support, he has increased the number of goats he can sell in each 3-month cycle from 20 to 70 goats and this additional income has enabled him to expand and diversify his other income generating activities.

In addition to trading goats, Abubakar operates a small self-financed mill (to process grains, primarily food rations), farms vegetables, trades charcoal, owns a retail shop and also owns livestock (cattle and camels) being cared for by clan/kin in another area. This gives him an estimated monthly income of 24,000 ETB (approximately £300). While he does not own land to farm, he rents land from a local landowner and grows onions with three other investors. In 2023 his onion crop was lost due to flooding and Abubakar suffered financial losses of around 1million ETB (£15,600) but he plans to invest in onions the next season.

Challenges such as the decrease in rations during 2023 which impacted his feed mill business and flooding which caused financial loss in his farming business continue to affect Abubakar. SHARPE's support has enabled him to strengthen his businesses and improved his ability to recover from shocks and his multiple income streams have given him the adaptive capacity to respond to the changing conditions.

⁵⁶ Reach (March 2024) The Impact of Drought and Climate-Related Shocks on Livelihood Practices in Somali (Factsheet) https://repository.impact-initiatives.org/document/reach/0579137a/REACH_ETH_Factsheet_SomaliLivelihoods_1504.pdf

UK International Climate Finance (ICF)⁵⁷: Contribution to adaptation and mitigation

International Climate Finance (ICF) is Official Development Assistance (ODA) from the UK allocated to support developing countries to reduce poverty and respond to and mitigate the challenges caused by climate change and environmental degradation.⁵⁸ The purpose of these investments is to help developing countries:

- Adapt and build resilience to the current and future effects of climate change
- Pursue low-carbon economic growth and development
- Protect, restore and sustainably manage nature
- Accelerate the clean energy transition.

ICF contribution to humanitarian programming is based on the ranking of the most climate-vulnerable countries according to the ND-GAIN index, which measures the climate vulnerability of countries based on a number of variables. Ethiopia currently ranks 163 out of the 185 ranked countries, just falling outside the 10% of countries most vulnerable to climate change.⁵⁹ Countries within the 10% receive a fixed proportion of 30% ICF. However, reflecting conflict and climate related drought and flooding, Ethiopia's rank has been falling since 2020 – if these conditions persist it may in the future fall within the 10% criteria.

While SHARPE is not funded by ICF, this study has considered whether the results meet the criteria. ICF results are reported for 11 KPIs and 4 TA KPIs.⁶⁰ Two of the KPIs are of direct relevance to SHARPE – ICF KPI 1 (adaptation) and ICF KPI 2.1 (clean energy-mitigation). ICF KPI 2.1 complements ICF KPI 1, which provides a population count of those supported to cope with the effects of climate change as a result of ICF programmes.

UK ICF KPI 1 – Adaptation

ICF KPI.1 Number of people supported to better adapt to the effects of climate change (as a result of ICF). This output indicator counts the number of people (total direct and total indirect beneficiaries) whom ICF programmes have supported to prepare and deal with the effects of climate change, including long-term changes in weather patterns and the increasing frequency and severity of extreme weather events.⁶¹



57 UK International Climate Finance (ICF) has four priorities: clean energy, nature for climate and people, adaptation and resilience, and sustainable cities, infrastructure and transport. For reporting, ICF uses a Key Performance Indicator (KPI) framework which includes 17 KPIs. <https://www.gov.uk/government/publications/uk-international-climate-finance-strategy>

58 The ICF portfolio is delivered jointly across four UK Government Departments: the Foreign Commonwealth & Development Office (FCDO); the Department for Energy Security and Net Zero (DESNZ formerly the Department for Business, Energy and Industrial Strategy); the Department for Science, Innovation and Technology (DSIT) and the Department for Environment, Food & Rural Affairs (Defra).

59 Countries in the region within the 10% include Eritrea, Somalia and Sudan (insufficient data available for South Sudan).

60 <https://www.gov.uk/government/publications/uk-international-climate-finance-strategy>

61 It particularly targets SDG 13.1: strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries.

KPI 1 is an output indicator that measures the **reach** of UK ICF climate change adaptation programmes. This KPI counts **the total number of people supported by ICF to be better able to adapt to the effects of climate change** but does not measure whether this support was successful in reducing people's vulnerability to effects of climate change. According to the KPI guidance, this should be assessed separately. Programmes that contribute to the UK ICF KPI 1 also support the achievement of Sustainable Development Goal (SDG) target 13.1 on strengthening resilience and adaptive capacity to climate-related hazards, as well as contributing towards SDG 2 on food security and SDG 6 on access to water.

Climate change adaptation is defined as 'the process of adjustment to actual or expected climate and its effects. In human systems, adaptation seeks to moderate or avoid harm or exploit beneficial opportunities. In some natural systems, human intervention may facilitate adjustment to expected climate and its effects.'⁶² To meet the KPI criteria, the programme is expected to reduce people's vulnerability to the effects of climate change by giving them **resources, tools, or skills to better cope with changing weather patterns**, including more unpredictable and extreme weather, as well as long-term, more incremental changes.

SHARPE has contributed to the resilience of both host and refugee communities. At the impact level SHARPE measures 196,045 people who benefitted from the programme at the impact level i.e. improved business and/or household income through either increased income, cost savings, a new or better job and/or increase in asset value. **Considering the ICF criteria for adaptation all SHARPE impact level beneficiaries can be considered to have stronger ability to adapt to climate change as a result of SHARPE since they have resources, tools or skills to better cope with changing weather patterns.** However, a sub-set (14,267) of these 196,045 people benefitted through access to renewable energy, and since people having improved access to energy should be reported against KPI 2.1 (below) and not against KPI 1 (according to the methodology guide for UK ICF indicators) SHARPE can legitimately claim to have supported 181,778 to better adapt to climate change.

Table 1: SHARPE contribution to UK KPI 1

Number of people supported to better adapt	181,778
Male	117,573 (65%)
Female	64,199 (35%)

UK ICF KPI 2.1 – Clean energy (Mitigation)

ICF KPI.2.1: Number of people and social institutions with improved access to clean energy (as a result of ICF).⁶³

ICF KPI 2.1 is an output indicator that counts the number of people who have improved access to clean energy as a result of an ICF intervention. UK ICF KPI 2.1 directly contributes to Sustainable Development Goal 7 (affordable and clean energy) and SDG 3 (health and wellbeing), reducing carbon emissions and improving health. Clean energy access refers to:

- New household connections to off-grid renewable energy sources such as micro or mini-grids, or solar home systems
- Households that use more efficient cookstoves that materially improve energy efficiency of essential services daily (e.g. air quality, reduced fuel consumption, fuel savings or improved family income)
- Solar lamps and/or solar lanterns with or without chargers

⁶² ICF, KPI 1 Methodology Note, February 2023

⁶³ ICF, KP2 Methodology Note, February 2023

UK ICF KP2.1 criteria includes interventions such as the installation of solar technology; green mini-grids for local power supply in rural areas; clean cook stoves that reduce people's dependence on firewood; and interventions that support improvements in energy efficiency. This is directly relevant to three SHARPE interventions on solar energy – home solar kits; productive use solar kits; and energy kiosks. These three interventions all work with the private sector to make quality solar supplies available and affordable for both refugees and remote host communities living in off-grid situations.

As of October 2024, through the SHARPE programme, **2,192 home solar energy kits** had been sold in the Somali region, improving clean energy access for 11,836 individuals⁶⁴. These solar kits power lighting (3 bulbs), phone charging and a torch and are paid using the PAY-GO system of monthly instalments, paid through Hello Cash and other mobile money platforms. All of these kits are in use (the company, Hello Solar, monitors monthly payments and knows the status of all the kits they sell).

Households purchasing a home solar energy kit can be calculated alongside the 2,430 people that benefit from having access to energy kiosks - 14 energy kiosk operators who collectively supply solar energy services (phone charging) to 450 households.⁶⁵

This takes the **SHARPE contribution to UK ICF KPI 2.1 to 14,266 people – 8,312 men (58%) and 5,954 women (42%)**



64 By the end of the programme 2,500 home solar kits will have been sold, benefitting 13,500 individuals.

65 Using a household multiplier of 5.4, as per ICF guidelines

Putting Climate Risk and Adaptation at the Centre of Programmes

This assessment found that SHARPE, a market systems development programme with multiple interventions working in multiple refugee hosting regions, is making a meaningful contribution to the climate resilience of refugee and host communities. However, the findings also suggest that if climate change had been integrated into the SHARPE programme from the beginning the results would be greater. There were missed opportunities to contribute to climate adaptation and mitigation. With the modification of some elements of these interventions, for example climate proofing infrastructure such as poultry and goat sheds, these contributions can be improved and strengthened.

It is crucial for vulnerable refugee communities, not only in Ethiopia but in other countries and regions at high climate risk, that climate resilience moves to the centre of programme design, planning and implementation. Over three-quarters of refugees globally (15.7 million in 2020) are in a protracted situation and new approaches are needed to drive the transition to sustainable development AND sustainable adaptation.⁶⁶ Anticipating climate-related risks holistically is key in steering humanitarian responses away from reactive towards proactive and resilient developmental paths.⁶⁷

A significant barrier to programmes increasing their climate contribution is that climate change and its related risks are not highlighted or effectively considered or integrated into programme design and planning. Climate risk is not considered as a primary driver for shaping economic development strategies, highlighted in Theories of Change, and prioritised in programme actions, but often only briefly mentioned in a background statement and actions for adaptation are ‘add-ons’ that are not adequately budgeted. As a result opportunities continue to be missed for strengthening adaptation and lowering emissions (e.g. integration of clean energy into programme targets).

To address this, a key step would be the use of a climate lens during programme design that would put to the forefront the risks, vulnerabilities and impacts related to climate variability and change. The programme context and actions should demonstrate a clear and direct link between the identified risks, vulnerabilities and impacts and the specific project activities. By using a climate lens during programme design, for example, business and services can be recognised as contributing to climate change adaptation in a number of ways e.g. by diversifying income opportunities for communities that cannot continue their traditional way of life as a result of climate change (such as is occurring in areas experiencing increasing climate risk), but also by supporting the development, distribution or adoption of new technologies to better deal with climate change.

For these barriers to fall, there has to be a shift from climate tweaked business-as-usual’ to a demand that climate risk be placed at the centre of sustainable development and adaptation. National economic development plans and international donor strategies are moving climate risk from the margins to the centre, but approaches (and mindsets) are struggling with the changes needed for planning for the ‘now’ with an eye to the long-term.

Key recommendations for all programmes seeking to address the longer-term livelihoods and prospects for economic integration of vulnerable refugee communities:

1. Conduct a detailed climate risk assessment during both scoping and inception phases
2. Integrate a climate lens into programme design [from the SHARPE perspective this would include adding a climate assessment to intervention concept notes and partnership agreements]
3. Include relevant UK ICF KPIs in programme log-frame to monitor contribution from the outset

66 https://www.unhcr.org/neu/wp-content/uploads/sites/15/2021/08/10-facts-about-refugees-2021-ENG_Vi2.pdf

67 UNHCR (2023) East and Horn of Africa and Great Lakes Region. Climate Action Plan (2023-2028) <https://data.unhcr.org/en/documents/details/104399>

Conclusions

Refugee-hosting sites, and refugee camps, are often located in marginal areas that are climatic ‘hotspots,’ intensified in some areas by refugee-host community conflict, conflict which is often driven by competition over natural resources. This is certainly the case in Ethiopia with more than half the refugee population housed in the Somali and Gambella regions, which are predicted to become hotter, drier, wetter with more frequent extreme climate events and greater unpredictability. Climate change is a threat multiplier for refugee and host communities undermining climate-sensitive livelihoods and destroying assets and resources.

Humanitarian assistance is at a crossroads, recognizing that it must turn more of its attention to supporting the self-reliance of protracted refugees, but hampered by diminishing resources and the lack of experience and expertise needed to strengthen and support the private sector and regional and local market systems which are essential for sustainable development. Given the protracted nature of many of these refugee settings, change change considerations need to be integrated into any intervention that seeks to improve livelihoods and self-reliance sustainably.

There are programmes such as SHARPE that are working in the refugee-market systems development space, seeking to create the conditions for greater economic inclusion of refugees and with the potential to support climate change adaptation. These programmes can provide evidence that a more systemic market systems development approach can succeed and, as in the case of SHARPE, be a model for sustainable development and sustainable adaptation.

The assessment found that even though climate risk was not integrated into its initial planning, SHARPE’s interventions were making significant contributions to climate adaptation and mitigation. Using the UK ICF criteria SHARPE has supported more than 130,000 people to increase their incomes and, as a result, are now better able to adapt to climate risk. In addition, through the SHARPE programme, more than 11,000 people now have access to clean energy. These achievements are notable given the relatively short period of implementation of the SHARPE programme through challenging conditions – COVID, extended drought, floods and inter-communal conflict.

Despite impressive achievements in increasing the resilience and adaptive capacity of both host and refugee communities, the assessment also found that **SHARPE could do even better** by making evidence-supported changes in some interventions. This could be facilitated by a more comprehensive approach (e.g. improved infrastructure combined with solar energy and financial services) to further address climate risk and strengthen resilient livelihoods in refugee hosting landscapes that are on the frontline of climate change. Climate risk was only one of many criteria included in initial assessments and intervention design – this could have been strengthened to ensure climate risk was a primary consideration when designing interventions.

Looking to the future, climate risk should be integrated from the beginning of programme planning for all climate-sensitive sectors. This is particularly the case for programmes seeking to improve the livelihoods, self-reliance and economic inclusion of refugees living in climate vulnerable, marginal regions. To be sustainable these programmes have to promote climate resilience livelihoods and support climate-sensitive economic inclusion.

The evidence from SHARPE that programmes currently working in the humanitarian-development space can make significant contributions to climate resilience should encourage other development programmes to assess if and what they are contributing to climate adaptation and mitigation, and to then identify how to do better. In situations of protracted displacement, considering climate risk in investments aiming at fostering self-reliance becomes even more pivotal as it holds the potential to reduce the critical need for on-going assistance over the mid-to long-term.

Annex 1: Glossary

Climate glossary – terms and definitions	
Term	Definition
Adaptation	Actions taken to moderate, cope with, or take advantage of actual or expected change in climate conditions with the objectives to: • Protect natural and human systems against the actual and anticipated harmful effects of climate change. • Exploit any opportunities they may generate. • Ensure the sustainability of investment and development interventions despite current climate risks and potentially increasingly difficult climatic conditions.
Adaptive capacity	The ability of a system to adjust in response to projected or actual changes in climate (including taking advantage of opportunities or coping with the consequences of a climate shock or stress).
Carbon sequestration	Process by which atmospheric carbon dioxide is taken up by plants through photosynthesis and stored as carbon in biomass and soils.
Climate	Often defined as the weather averaged over time (typically 30 years).
Climate change	Change in climate over time, whether due to natural variability or as a result of human activity. The United Nations Framework Convention on Climate Change (UNFCCC) defines climate change as ‘a change of climate which is attributed directly or indirectly to human activity that alters the composition of the global atmosphere and which is in addition to natural climate variability observed over comparable time periods.’
Climate change impacts	The effects of climate change (e.g. increasing temperatures and change in seasonal patterns) and extremes (e.g. heat waves, heavy rainfall, severe storms) on human systems and the environment. The impacts of climate change include: • Physical impacts – floods, droughts and sea level rise. • Environmental – impacts on ecosystems such as loss of biodiversity, water scarcity, and land degradation. • Impacts on human systems – socio-economic effects (e.g. loss of life and livelihood, food insecurity), and infrastructure consequences.
Climate hazard	Any climate-related physical event (e.g. cyclone), trend (e.g. increasing temperatures in dry season) or impact (e.g. flood, sea level rise), which has the potential to cause loss of, and damage to people, physical assets, provision of services, and ecosystems. Hazards (as well as climate change impacts) can be divided into slow onset (like drought and insect infestations), and rapid onset (like floods) events.
Climate projections	Simulation of possible future state of earth’s climate (e.g. in 2050 and 2100) using climate models.
Climate-proofing	Enhancing resilience to climate change by analysing the risks posed by climate change to activities, stakeholders, and results, then modifying and/or adjusting project designs or implementation plans to mitigate these risks. For example, building water pumps above predicted flood heights in order to ensure the availability of safe water over the longer-term.
Climate resilience	Enhancing resilience to climate change by improving preparedness (early warning and decision support systems, response and relief), risk reduction (robust agriculture, water resources management, livelihood support) and risk mitigation (safety nets, agriculture insurance, risk financing facility).

Climate-resilient development (CRD)	A process of implementing greenhouse gas mitigation and adaptation options to support sustainable development for all. (IPCC Sixth Assessment Report). CRD ensures the ability to cope with current climate and its impact and to adapt to future climate change, by preserving development gains and minimizing damages.
Climate Rio Markers	Climate markers provide an indication of the policy goals of the donors that contribute to an activity. An activity is assessed to determine if climate change is an objective. • A score of 'principle' (2) is assigned when climate change is fundamental in the design of the activity and is an explicit objective of the activity. • A score of 'significant' (1) is assigned when climate change is important but is not one of the principal reasons for undertaking the activity. • A score of 'not targeted' (0) is assigned when the activity has been screened and found not to target climate change.
Climate risk	The potential for climate change to create adverse consequences for human or ecological systems. This includes impacts on lives, livelihoods, health and wellbeing, economic, social and cultural assets and investments, infrastructure, services provision, ecosystems and species
Climate risk management	Seeks to anticipate, avoid and prevent climate risks. Examples: Improving the national meteorological services to provide better weather forecasts that reach farmers, enhanced early warning systems, crop and livestock insurance so farmers are protected against losses; contingency planning, contingent financing and social protections; and technologies and practices that can deal with extreme conditions.
Climate-smart development	Adding up the benefits of actions that help build prosperity, end poverty and combat climate change. The underlying concept is that the right policy choices can deliver significant benefits to lives, jobs, crops, energy, and GDP – as well as emissions reductions to combat climate change. (Note: climate resilient development is now more frequently used than climate-smart development.)
Climate variability	Refers to variations in the mean state of climate on all temporal and spatial scales beyond that of individual weather events. Examples of climate variability include extended droughts, floods, and conditions that result from periodic El Niño and La Niña events.
Disaster risk	The potential loss of life, injury, or destroyed or damaged assets which could occur to a system, society or a community in a specific period of time. Disasters can be caused by natural, man-made and technological hazards, as well as various factors that influence the exposure and vulnerability of a community.
Exposure	The presence of people, livelihoods, environmental services and resources, infrastructure, or economic, social or cultural assets in places that could be adversely affected.
Maladaptation	Outcome of efforts to adapt that either result in increased vulnerability to climate change or undermine the ability to adapt in the future.
Mitigation	Actions that aim to reduce emissions, directly or indirectly, by avoiding or capturing greenhouse gases (GHG) before it is emitted to the atmosphere or sequestering GHG already in the atmosphere by enhancing 'sinks' such as forest.

Sensitivity	Sensitivity is the degree to which a system is affected, either adversely or beneficially, by climate variability or change. The effect may be direct (e.g., a change in crop yield in response to a change in the mean, range, or variability of temperature) or indirect (e.g., damages caused by an increase in the frequency of coastal flooding due to sea level rise).
Vulnerability	To the impacts of climate change – the propensity or predisposition to be adversely affected. It is used to describe socio-economic, physical and environmental factors, which determine the sensitivity/susceptibility of a country, city, village, community or individual to the impact of climate change (e.g. change in seasonal patterns) and/or hazard (e.g. flood).
Weather	Describes atmospheric conditions at a particular place and at a certain time in terms of temperature, precipitation, wind speed, humidity, atmospheric pressure, and cloudiness.





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